

Possible Answers :

10.75 to 10.85

Industry 4.0

Section Id :	64065345318
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065396848
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 105 Question Id : 640653676883 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532266772. ✓ YES

6406532266773. ✗ NO

Sub-Section Number : 2
Sub-Section Id : 64065396849
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 106 Question Id : 640653676884 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Match Col A with Col B for landmark outcome of industrial era

Col A (Industrial Era)	Col B (Major Outcome)
1.0	Digitization
2.0	CNC
3.0	Steam Engine
4.0	Mass production

Options :

6406532266774. ✗ 1.0:CNC, 2.0:Mass production, 3.0: Digitization, 4.0:Steam Engine

6406532266775. ✗ 1.0:Steam Engine, 2.0:mass production, 3.0: digitization, 4.0:CNC

6406532266776. ✗ 1.0:Mass production, 2.0:Steam Engine, 3.0: digitization, 4.0:CNC

6406532266777. ✓ 1.0:Steam Engine, 2.0:mass production, 3.0: CNC, 4.0: Digitization

Sub-Section Number : 3
Sub-Section Id : 64065396850
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 107 Question Id : 640653676885 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Multiple Choice Question
Identify the Industry 4.0 version of transportation.

Options :

- 6406532266778. ✖ IC engine
- 6406532266779. ✖ Electric car
- 6406532266780. ✔ Hyperloop

Question Number : 108 Question Id : 640653676889 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Multiple Choice Question
The Production rate of an assembly line is determined by its

Options :

- 6406532266792. ✔ Slowest station
- 6406532266793. ✖ Fastest station
- 6406532266794. ✖ Idle station
- 6406532266795. ✖ None of these

Sub-Section Id :

64065396851

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 109 Question Id : 640653676886 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question
Select the layout(s) which require(s) general supervision:

Options :

6406532266781. ✓ Product layout
6406532266782. ✖ Process layout
6406532266783. ✓ Group layout
6406532266784. ✓ Fixed position layout

Sub-Section Number :

5

Sub-Section Id :

64065396852

Question Shuffling Allowed :

Yes

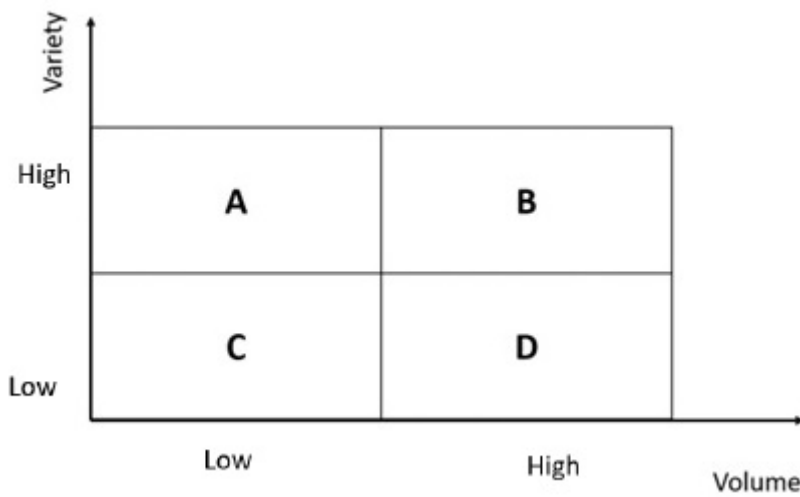
Is Section Default? :

null

Question Number : 110 Question Id : 640653676887 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Multiple Choice Question
Select the appropriate layout for Quadrants A, B, D in the following figure:



Options :

6406532266785. ✖ A: Group layout, B: Product layout , D: Process layout

6406532266786. ✔ A: Process layout, B: Group layout , D: Product layout

6406532266787. ✖ A: Product layout, B: Process layout , D: Group layout

Sub-Section Number :

6

Sub-Section Id :

64065396853

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 111 Question Id : 640653676888 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is an advantage of Product Layout?

Options :

6406532266788. ✔ Small in-process inventory

6406532266789. ✖ It can accommodate high variety of products

6406532266790. ✖ High utilization of machines

6406532266791. ✖ Low investment

Sub-Section Number : 7
Sub-Section Id : 64065396854
Question Shuffling Allowed : Yes
Is Section Default? : null

Question Number : 112 Question Id : 640653676890 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406532266796. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region
6406532266797. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity
6406532266798. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point
6406532266799. ✖ Optimal solution in a facility location problem using Euclidean method matches the center of gravity in not of the cases

Question Number : 113 Question Id : 640653676891 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the examples for Offshoring from the list below:

Options :

6406532266800. ✔ Shell, an oil company based in Netherlands, setting up a data science team in

India.

6406532266801. ✖ Western Australia’s electricity network agency assigns Tata Consultancy Services at Pune, to support it with an overhaul of its legacy IT systems, aimed at delivering cheaper and more reliable power.

6406532266802. ✔ Target Corporation, a retail giant based in US, setting up Supply Chain Optimization team at Bengaluru.

6406532266803. ✖ Tata Motors assigns the manufacturing of its braking system to TACO group.

Sub-Section Number :	8
Sub-Section Id :	64065396855
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653676892 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (114 to 115)

Question Label : Comprehension

A factory works for two 8-hour shifts each day and produces a total of 30,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. Answer the given subquestions:

Sub questions

Question Number : 114 Question Id : 640653676893 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Correct Marks : 1

Question Label : Short Answer Question

What is the cycle time (in seconds) at any workstations (round your answer to one decimal).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.7 to 2.0

Question Number : 115 Question Id : 640653676894 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the theoretical minimum number of workstations required?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Sub-Section Number : 9

Sub-Section Id : 64065396856

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653676895 Question Type : COMPREHENSION Sub Question Shuffling

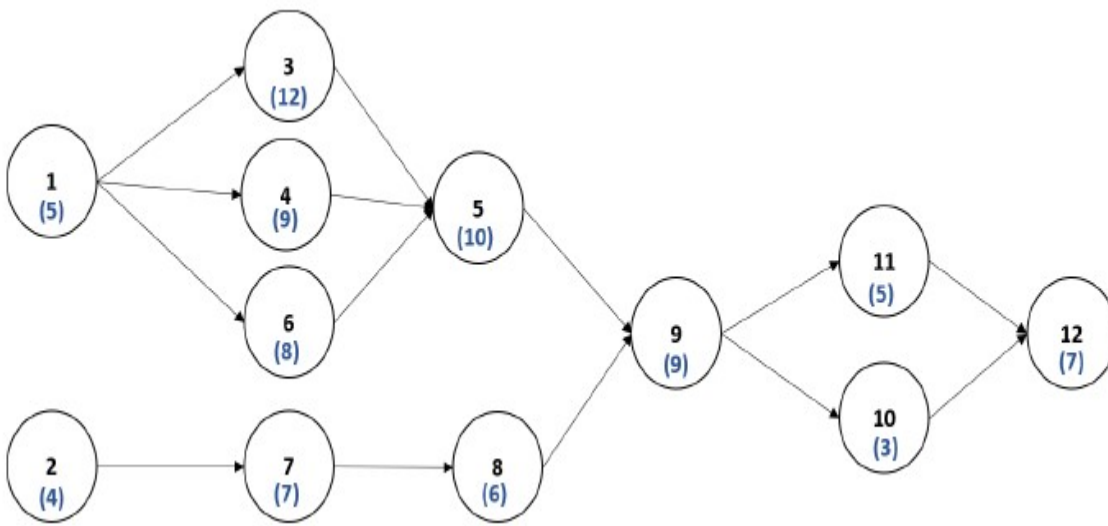
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (116 to 120)

Question Label : Comprehension

Consider the following precedence diagram. It has 12 activities. The numbers in blue and withing the braces denote the respective activity times in minute.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 116 Question Id : 640653676896 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the minimum cycle time of the system (in minute)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 117 Question Id : 640653676897 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the maximum cycle time of the system (in minute)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

85

Question Number : 118 Question Id : 640653676898 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

The factory works for one 8-hour shifts each day. Is it possible to achieve an output of 32 units/day,

Options :

6406532266808. ✓ Yes

6406532266809. ✗ No

Question Number : 119 Question Id : 640653676899 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the balance delay of the following layout (round it to two decimals)

(1,2) -> (3) -> (4) -> (5) -> (6,7) -> (8,9) -> (10,11,12)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText